Our Magical Statistics Adventure

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Research Question

We examine how healthcare expenses are associated with patient age, gender, and healthcare utilization, measured as the number of encounters.

Our main research question is:

How are healthcare expenses associated with age, gender, and healthcare utilization (number of encounters)?

We also explore three related sub-questions:

- How do healthcare expenses vary across age groups?
- Are there differences in healthcare expenses between male and female patients?
- Is the number of healthcare encounters associated with healthcare expenses across genders?

Descriptive Statistics

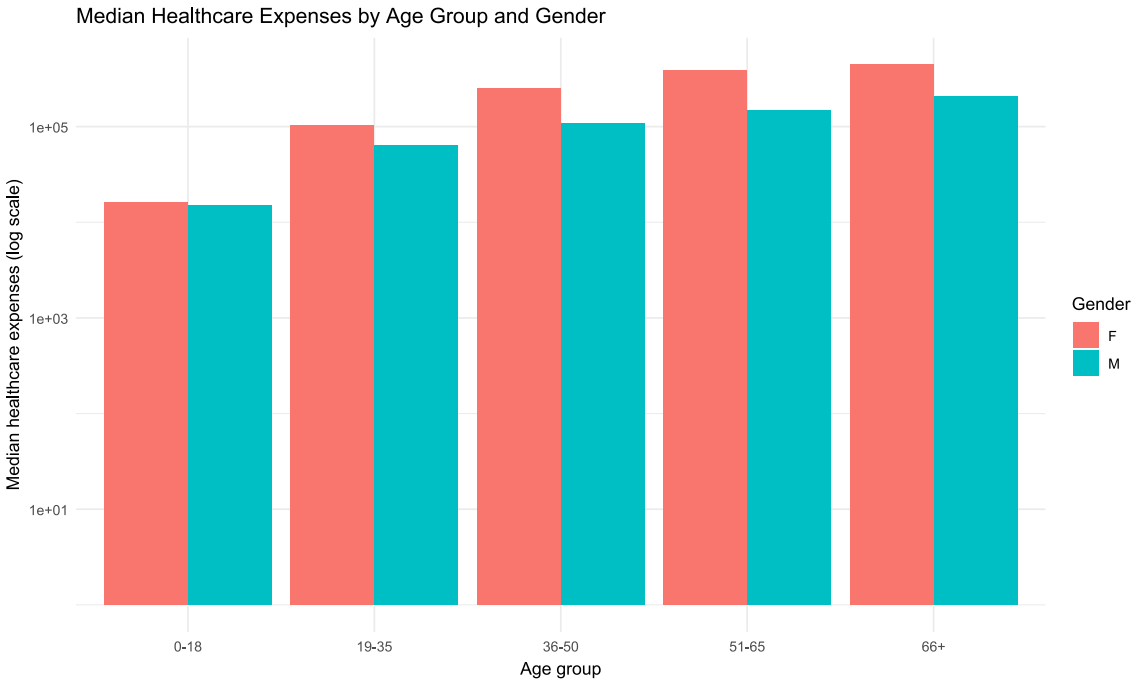
Characteristic	**Overall** N = 6,851 [1]	**F** N = 3,458 [1]	**M** N = 3,393 [1]	**p-value** [2]
age	42 (22, 62)	41 (21, 60)	43 (23, 64)	<0.001
n_encounters	96 (43, 180)	124 (46, 213)	80 (40, 141)	<0.001
healthcare_expenses	103,689 (22,444, 260,417)	152,023 (23,258, 421,262)	85,742 (21,343, 171,986)	<0.001

[1] Median (Q1, Q3)

[2] Wilcoxon rank sum test

- Approximately equal number of males and females
- Age is balanced between genders
- Encounters for women are higher
- Medical expenses for women are higher

Barchart of Expenses by Age and Gender



- Even though age and gender were evenly distributed, expenses for women are clearly higher.
- As age increases, so do expenses for both genders.

Scatterplot of Encounters to Expenses Compared Between Genders

```
`geom_smooth()` using formula = 'y ~ x'
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```

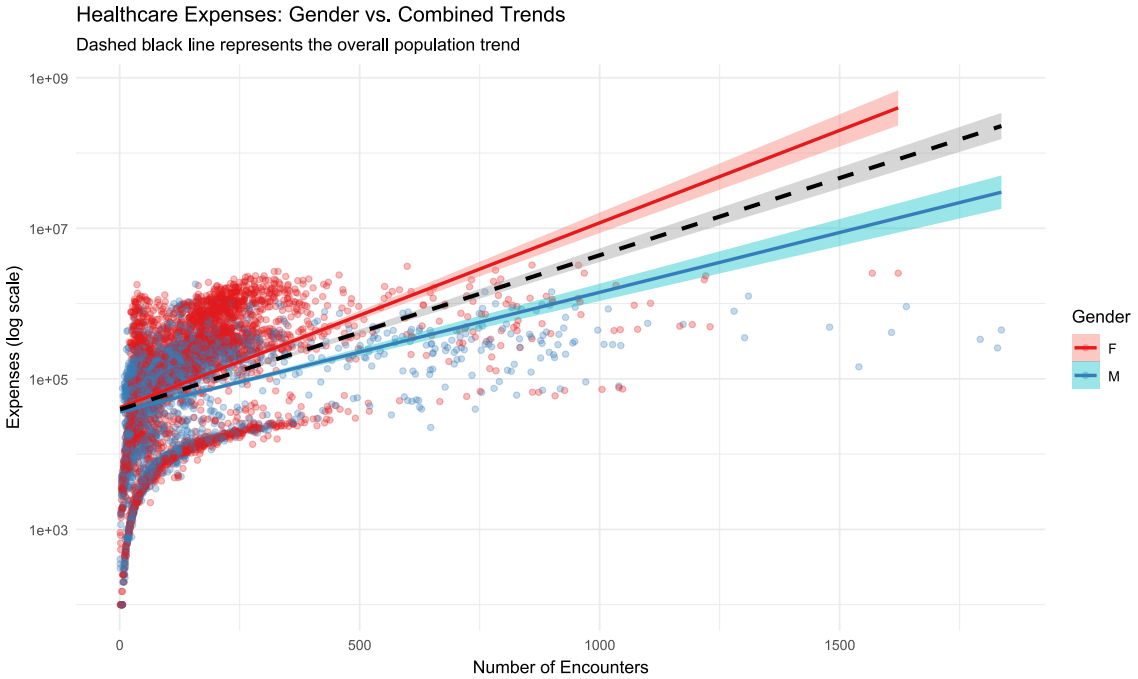

- Both genders have increased cost for increased encounters.
- The increase is steeper for women than men.

Linear Regression

```
# Since regression_results is a raw 'lm' object,
# we use tbl_regression to make it look pretty like the rest!
#| echo: false
library(gtsummary)
tar_read(regression_results) |>
  tbl_regression()
```

Characteristic	Beta	95% CI	p-value
age	0.04	0.04, 0.04	<0.001
gender			
F	—	—	
M	−0.60	−0.67, −0.54	<0.001
n_encounters	0.00	0.00, 0.00	<0.001

Abbreviation: CI = Confidence Interval

log(1 + healthcare_expenses) ~ age + gender + n_encounters

- Increasing age leads to increasing expenses

- Being male decreases expenses
- Increasing encounters increases expenses

All are significant

`Thank you!` 💖